CHAPTER 6

The logarithm function

S44, S47, S49, S50

Objectives

Use the properties of ln; solve equations and thresholds; differentiate and study logarithmic functions.

Course notes

Definition and properties

The exponential is continuous and strictly increasing on $\mathbb{R}$, and its range is $]0,+\infty[$. For every $x>0$, there is a unique real number y such that $e^y = x$: we denote it by $\ln x$.

$$\ln(e^y) = y \quad (y \in \mathbb{R}), \qquad e^{\ln x} = x \quad (x>0), \qquad \ln 1 = 0, \quad \ln e = 1.$$

For $a, b > 0$ and $n \in \mathbb{Z}$,

$$\ln(ab) = \ln a + \ln b, \quad \ln\left(\frac{a}{b}\right) = \ln a - \ln b, \quad \ln(a^n) = n \ln a, \quad \ln\sqrt{a} = \tfrac{1}{2}\ln a.$$

Proof of the product rule. $e^{\ln a + \ln b} = e^{\ln a} e^{\ln b} = ab$; the uniqueness of the preimage gives the formula. There is no analogous formula for $\ln(a+b)$.

Derivative, variations and limits

We accept that ln is differentiable on $]0,+\infty[$. Differentiating $e^{\ln x} = x$ gives $e^{\ln x}(\ln x)' = 1$; hence $(\ln x)' = \frac{1}{x} > 0$. ln is strictly increasing and concave, because its second derivative is $-\frac{1}{x^2}$.

$$\lim_{x\to 0^+} \ln x = -\infty, \qquad \lim_{x\to+\infty} \ln x = +\infty.$$

Its curve is the reflection of that of the exponential in the line $y = x$. The line $x = 0$ is a vertical asymptote. Its tangent at 1 gives $\ln x \leq x - 1$ for $x > 0$.

Composition and comparison of growth rates

If $u > 0$ is differentiable, $(\ln u)' = \frac{u'}{u}$. For every integer $m \geq 1$,

$$\lim_{x\to+\infty} \frac{\ln x}{x^m} = 0, \qquad \lim_{x\to 0^+} x^m \ln x = 0.$$

Proof at 0. Set $t = -\ln x$; as $x \to 0^+$, $t \to +\infty$ and $x \ln x = -te^{-t} \to 0$ by comparison of growth rates. For $m \geq 1$, the same reasoning gives $-te^{-mt} \to 0$. At $+\infty$, set $t = \ln x$ to write $\frac{\ln x}{x^m} = \frac{t}{e^{mt}} \to 0$.

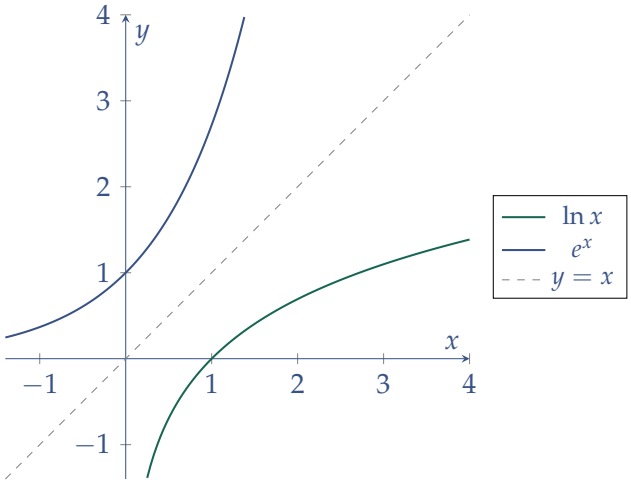
Equations and thresholds

For $A, B > 0$, $\ln A = \ln B \Leftrightarrow A = B$ and $\ln A \leq \ln B \Leftrightarrow A \leq B$. Always start with the domain.

Example. $\ln(x-1)+\ln(x+1)=\ln 8$ requires $x>1$. The equation becomes $x^2-1=8$, so $x=3$ only.

Threshold. $0.8^n \leq 0.1$ is equivalent to $n\ln 0.8 \leq \ln 0.1$, then $n \geq \frac{\ln 0.1}{\ln 0.8} \simeq 10.319$. The first integer is 11.

Example of a study. For $f(x)=\frac{\ln x}{x}$, $f'=\frac{1-\ln x}{x^2}$; f increases on $]0,e]$, decreases on $[e,+\infty[$, and its maximum is $\frac{1}{e}$. Its limit at 0^+ is $-\infty$, its limit at $+\infty$ is 0.



Research activity

Controlling a calculation error — 30 min.

A calculation method produces an initial error $e_0=5$. At each step, the error is multiplied by 0.7. Denote by e_n the error after n steps, for $n \in \mathbb{N}$.

1. Express e_n in terms of n.
2. Using the logarithm, determine the smallest integer n such that $e_n \leq 10^{-4}$.
3. Using a calculator, calculate the error at the value found and at the previous value to check that it is minimal.
4. We now require an error of at most 10^{-6}. Determine the new minimum number of steps. Does dividing the tolerance by 100 multiply the number of steps by 100?

Progressive exercises

E1 · Practice. After determining the set on which it is defined, solve the equation $\ln(2x-1)=\ln(x+3)$.

E2 · Practice. Determine the real numbers x for which $\ln x+\ln(x-2)$ is defined, then solve $\ln x+\ln(x-2)=\ln 3$.

E3 · Practice. Consider $f(x)=\ln(x^2+1)$ and $g(x)=x\ln x$. State their domains of definition and calculate their derivatives.

E4 · Practice. Determine $\lim\limits_{x\to+\infty}\dfrac{\ln x}{\sqrt{x}}$ and $\lim\limits_{x\to 0^+} x^2\ln x$. Justify the comparison-of-growth-rates results used.

E5 · Practice. By studying a function, prove that $\ln x \leq x - 1$ for every real number $x > 0$. Do not use the concavity of the logarithm directly.

Textbook or past-paper work

Barbazo, p. 196, exercise 36. At home after S44: do the three functions. Write down the conditions of definition before solving the inequalities; justify the open endpoints of the intervals obtained.

Homework

Homework 6 — Study of a logarithmic function — 55 min.

Consider the function f defined on $]0;+\infty[$ by $f(x) = \frac{1+\ln x}{x}$. Its graph is denoted $\mathcal{C}_f$.

1. Determine the limits of f at the endpoints of its domain, study its sign, then draw up its table of variations.
2. Study the convexity of f and determine the coordinates of the point of inflection of $\mathcal{C}_f$.
3. Prove that the equation $f(x) = \frac{1}{2}$ has exactly two solutions. Give for each a justified bracketing using calculated values.
4. Using the maximum of f, prove that $1 + \ln x \leq x$ for all $x > 0$. State the case of equality.

Teaching notes and solutions

Teacher notes

S44 is placed before the first mock exam, but the logarithm is not part of its announced scope. Stress the conditions $a > 0, b > 0$ before combining logarithms. The differentiability of ln is accepted; the differentiation formula is proved. For a threshold with $0 < q < 1$, the negative sign of $\ln q$ reverses the inequality. Have students check the candidate value and the previous one.

Solutions to the activity

1. Each step multiplies the error by 0.7: $e_n = 5(0.7)^n$ for $n \geq 0$.

2. $e_n \leq 10^{-4}$ is equivalent to $(0.7)^n \leq 2 \times 10^{-5}$. Since ln is increasing, $n \ln 0.7 \leq \ln(2 \times 10^{-5})$. As $\ln 0.7 < 0$, we obtain
$$n \geq \frac{\ln(2 \times 10^{-5})}{\ln 0.7} \simeq 30.33512.$$
The first suitable integer is 31.

3. $e_{30} \simeq 1.1270 \times 10^{-4} > 10^{-4}$ and $e_{31} \simeq 7.8888 \times 10^{-5} < 10^{-4}$. The sequence of errors is decreasing, so this check confirms minimality.

4. For the target 10^{-6}, the same calculation gives $n \geq \frac{\ln(2\times10^{-7})}{\ln 0.7} \simeq 43.2465$, so $n = 44$. The real threshold increases by $\frac{\ln 100}{|\ln 0.7|} \simeq 12.91$, then we round up to the next integer. The number of steps is not multiplied by 100.

E1. We need $2x - 1 > 0$ and $x + 3 > 0$, so $x > \frac{1}{2}$. On this domain, the injectivity of ln gives $2x - 1 = x + 3$, that is $x = 4$. This value belongs to the domain; the solution set is $\{4\}$.

E2. We need $x > 2$. The product rule gives $\ln(x(x-2)) = \ln 3$, so $x^2 - 2x - 3 = 0$, that is $(x-3)(x+1) = 0$. Only $x = 3$ satisfies $x > 2$: the other root -1 is excluded.

E3. Since $x^2 + 1 > 0$, the first function is differentiable on $\mathbb{R}$ and its derivative is $\frac{2x}{x^2+1}$. On $]0, +\infty[$, the product rule gives $(x \ln x)' = \ln x + x(\frac{1}{x}) = 1 + \ln x$.

E4. For the first limit, set $t = \sqrt{x} \to +\infty$: $\frac{\ln x}{\sqrt{x}} = \frac{2 \ln t}{t} \to 0$. For the second, set $t = -\ln x \to +\infty$: $x^2 \ln x = -te^{-2t} \to 0$. Both limits use comparison of growth rates.

E5. For $x > 0$, set $g(x) = x - 1 - \ln x$. We have $g'(x) = \frac{x-1}{x}$, negative before 1, positive after. Its minimum is $g(1) = 0$, so $g(x) \geq 0$, which is equivalent to $\ln x \leq x - 1$.

Solutions to Homework 6

1. At 0^+, the numerator tends to $-\infty$ and the denominator to 0^+: $f \to -\infty$. At $+\infty$, $f = \frac{1}{x} + \frac{\ln x}{x} \to 0$. The sign is that of $1 + \ln x$: negative on $]0, e^{-1}[$, zero at e^{-1}, positive afterwards. By the quotient rule,
$$f'(x) = \frac{1 - (1 + \ln x)}{x^2} = -\frac{\ln x}{x^2}.$$
Thus f strictly increases up to 1, then strictly decreases; its maximum is $f(1) = 1$.

2. Differentiating $-(\ln x)x^{-2}$, we obtain $f''(x) = \frac{2 \ln x - 1}{x^3}$. The numerator changes sign, from negative to positive, at $x = \sqrt{e}$. So f is concave on $]0, \sqrt{e}]$, convex on $[\sqrt{e}, +\infty[$ and has the point of inflection $(\sqrt{e}, \frac{3}{2\sqrt{e}})$.

3. On $]0,1]$, continuity, strict increase, limit $-\infty$ and final value 1 give exactly one preimage of $\frac{1}{2}$. On $[1,+\infty[$, strict decrease from 1 towards 0 gives exactly one second preimage. The first is in $]e^{-1},\frac{1}{2}[$, because $f(e^{-1})=0$ and $f(\frac{1}{2})=2(1-\ln 2)\simeq 0.6137>\frac{1}{2}$. The second is in $]5,6[$, because $f(5)\simeq 0.5219>\frac{1}{2}>f(6)\simeq 0.4653$.

4. The maximum gives $\frac{1+\ln x}{x}\leq 1$. Since $x>0$, we can multiply without changing the direction: $1+\ln x\leq x$. Equality corresponds to the unique maximum of f, so to $x=1$.

Solutions to the textbook or past-paper work

The logarithm requires a strictly positive argument.

1. $3x-7>0\iff x>\frac{7}{3}$, so $D_f=]\frac{7}{3};+\infty[$.

2. $-x^2+4x-3=-(x-1)(x-3)>0$ exactly when $1<x<3$. Thus $D_g=]1;3[$.

3. We need simultaneously $x>0$ and $2-x>0$; their intersection gives $D_h=]0;2[$. The coefficient -3 placed in front of the second logarithm does not change its condition of definition.